

Principles of Economics

Problem Set 8

Universidad Carlos III de Madrid | Department of Economics

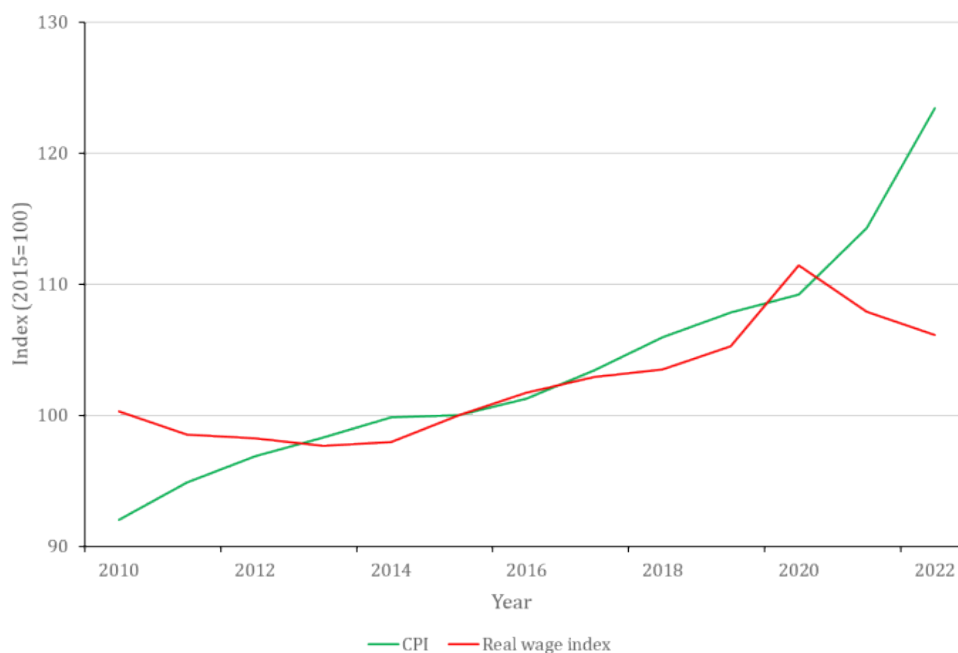
© 2026 Antonio Cabrales, Laurentiu Guinea Voinea, Ángel Hernando-Veciana, Bruno Pessoa and the Principios de Economía teaching team, Universidad Carlos III de Madrid. Licensed under CC BY-NC-SA 4.0, except where otherwise noted. Figures from CORE Econ, *The Economy 2.0*, under CC BY-NC-ND 4.0.

1. WARMING-UP QUESTIONS

1. In the model in this unit, the unemployed are no different from the employed (except for their bad luck). Imagine you are an employer, and one of the unemployed comes to you and promises to work at the same effort level as your current workers but at slightly lower wage. How would you reply?
2. Does your reply in the previous question help explain why unemployment must exist in a Nash equilibrium?
3. Download US data for the [Consumer Price Index (CPI)] (<https://tinyco.re/1753975>) and [real wages] (<https://tinyco.re/5349552>) (measured as weekly earnings) from the St Louis Federal Reserve website.

For this exercise, only use the data from 2010 (or January 2010) to the latest available year (or month).

- a. The monthly CPI data currently has the months in 1982–1984 as the base. Rebase the CPI data so that January 2015 is the base (the CPI for January 2015 should equal 100).
- b. Create an index for real wages with 2015 as the base year (the real wage for 2015 should equal 100).
- c. Use statistical software to make a line chart similar to Figure 1.9 for your rebased data from Questions 1 and 2. Discuss the similarities and differences between the charts.
- d. Discuss some criteria that you think would be important for choosing the base year for a data series.



4. SHIFTS IN THE WAGE-SETTING CURVE

For each of the following changes, provide a brief explanation of which direction the wage-setting curve shifts (upwards/leftwards or downwards/rightwards) and why, referring to the no-shirking wage and/or reservation wage.

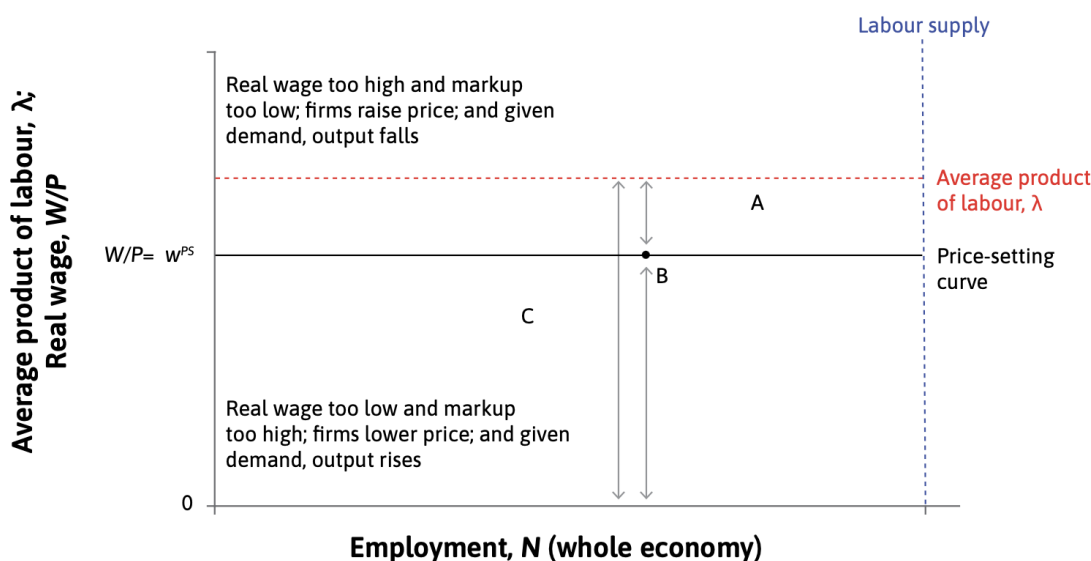
- Decrease in unemployment benefit
 - Decrease in social stigma attached to being unemployed
 - A decrease in the disutility of working
5. Suppose that some of the immigrants to the country decide to set up businesses, rather than become employees. Explain how you expect this to affect the wage-setting curve, the price-setting curve, and the labour market equilibrium.

2. MULTIPLE-CHOICE QUESTIONS

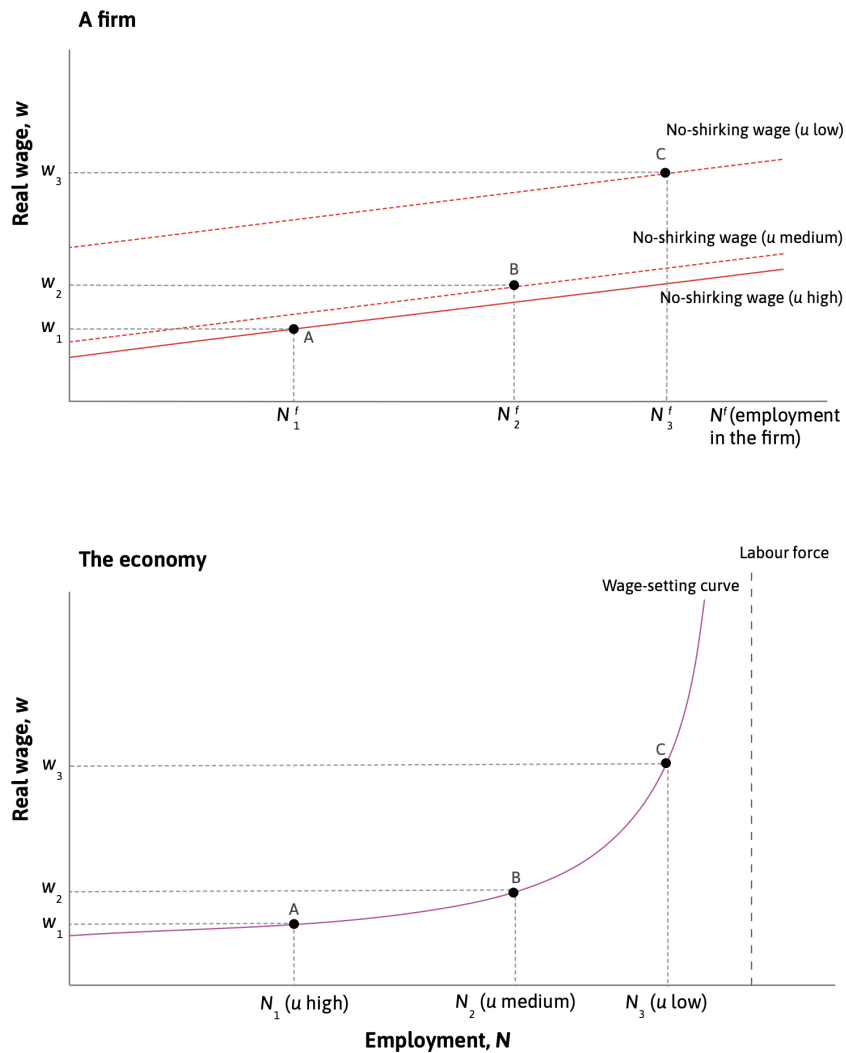
6. Read the following statements and choose the correct option(s).

- Participation rate = employed/labour force.
- Unemployment rate = unemployed/population of working age.
- Employment rate = employed/population of working age.
- Employment rate + unemployment rate = 1.

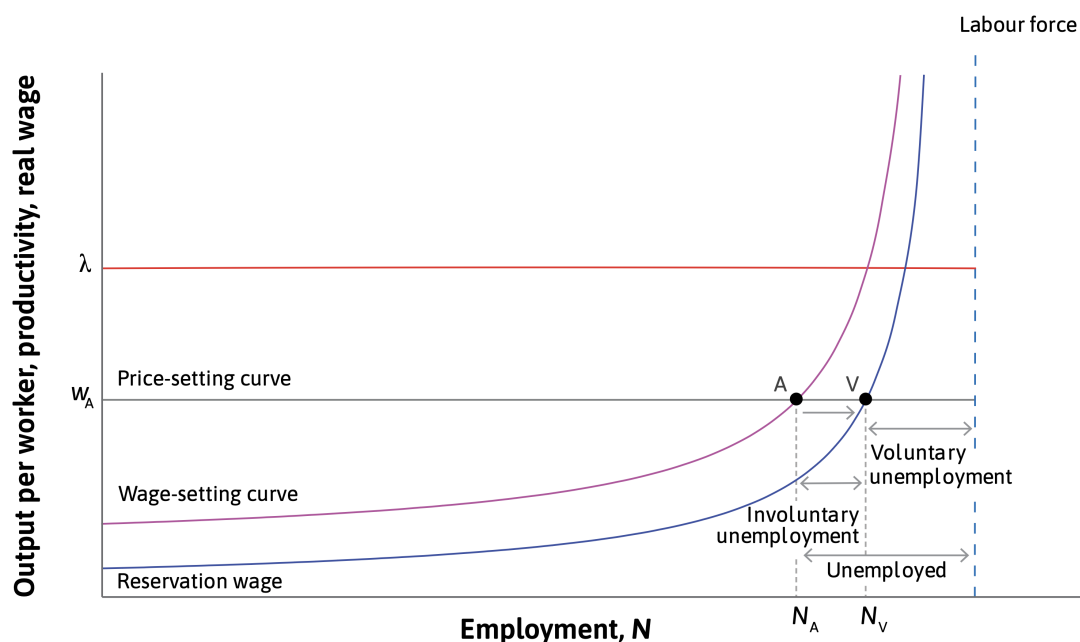
7. The following diagram depicts the price-setting curve. Based on this information, which of the following statements is correct?



- At point A, the markup is too high, and therefore the firm will raise its price. This leads to lower demand for the good and hence lower employment towards B.
 - At point C, the real wage is too low and the markup is too high. Therefore, the firm is able to increase profit by lowering prices and hiring more workers.
 - Higher competition implies a lower price-setting curve.
 - For any given markup, higher labour productivity implies a lower price-setting curve, which means a lower real wage.
8. The next figure shows the wage-setting curve and how it is derived from the no-shirking wage faced by individual firms. Based on this figure, read the statements and choose the correct option(s).
- An increase in the unemployment benefit would raise the no-shirking wage and shift the wage-setting curve downwards.
 - If the unemployment rate decreases (other firms' demand for labour increases), it would shift the no-shirking wage upwards and the wage-setting curve upwards.
 - If it becomes easier to monitor effort, the no-shirking wage decreases and the wage-setting curve shifts downwards.
 - A sudden drop in the working-age population (due, for example, to the retirement of the baby-boomer generation) would shift the no-shirking wage and wage-setting curve lower.



9. The next figure shows point A as the Nash equilibrium of the WS–PS model. If firms face a lower degree of competition in the market for goods and services, then:
- The price-setting curve shifts upwards.
 - The wage-setting curve shifts downwards.
 - The equilibrium real wage falls.
 - The unemployment level falls.



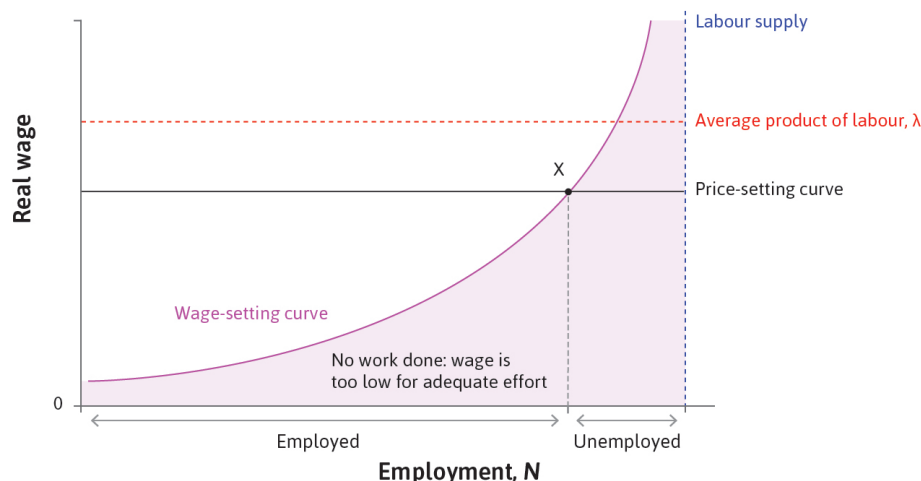
10. You are given the following data for the five countries. The numbers are in thousands. Of the five countries, which of the following statements is correct?

	Greece	Italy	Japan	UK	USA
Population of working age	9,213	52,058	110,780	52,812	253,538
Labour force	4,804	25,770	66,480	33,239	159,187
Employed	3,674	22,758	64,400	31,640	151,436

- UK has the smallest number of unemployed people.
 - USA has the highest participation rate.
 - Italy has the lowest employment rate.
 - Japan has the lowest unemployment rate.
11. Which of the following statements about the wage-setting curve is correct?
- The wage-setting curve depicts the workers' reservation wage for different levels of economy-wide employment.
 - At each point (U, w) on the wage-setting curve, the workers are choosing their best response effort level given the real wage w and unemployment rate U .
 - A lower unemployment rate shifts the wage-setting curve to the left.
 - An exodus of European workers due to Brexit would, *ceteris paribus*, result in a downward shift of the UK's wage-setting curve.
12. Which of the following statements about the price-setting curve is correct?
- The price-setting curve depicts the firms' profit-maximising price level for different levels of economy-wide employment.
 - Firms have to pay a higher real wage when the employment rate is higher. Therefore the price-setting curve is upward-sloping.

- c. At points below the price-setting curve, the firms are setting prices too high compared to their profit-maximising level.
- d. The falling European competition due to Brexit would result in the UK's price-setting curve shifting up.

13. The figure depicts the labour market model. *Ceteris paribus*, in which of the following cases would the equilibrium real wage level rise?



- a. Higher labour productivity.
- b. Lower competition in the output goods market.
- c. An exodus of immigrant workers.
- d. Higher unemployment benefit.

3. NUMERICAL PROBLEMS

14. Visit the ILO's website and use the ILOSTAT Database, (<https://tinyco.re/2173706>):

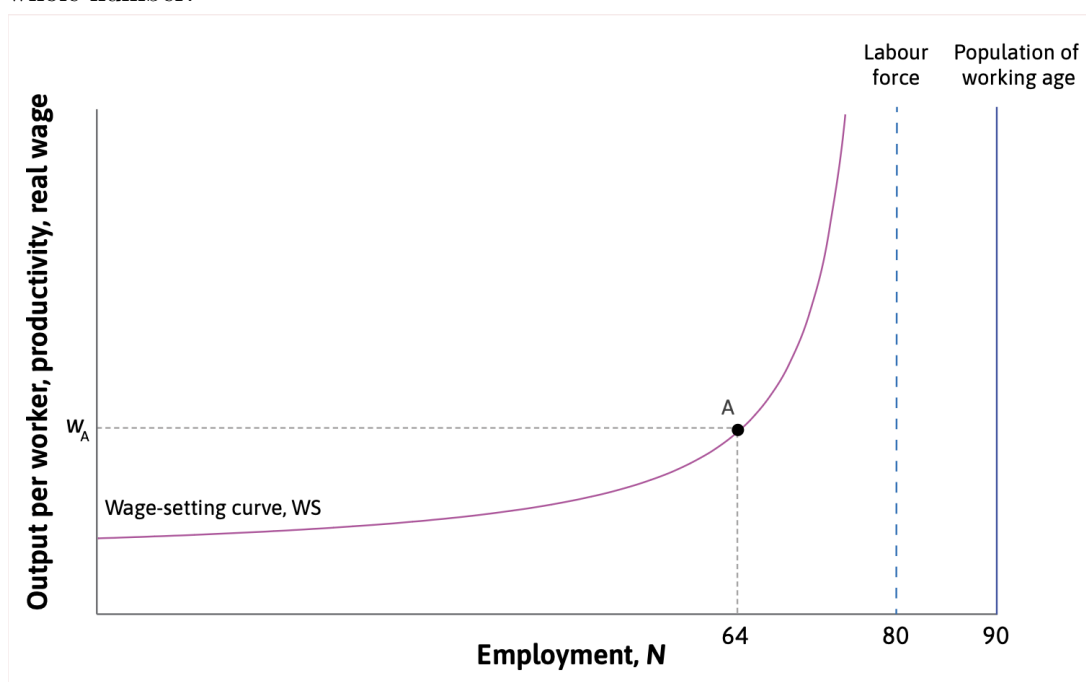
- a) to calculate the **average employment**, **unemployment**, and **participation rates** over the period 2000–2019, for two economies of your choice.
- b) Create appropriate chart(s) to display these statistics for the US, UK, and your chosen countries. Using your chart(s), describe the similarities and differences in these statistics between your chosen countries, the US, UK, and those in Figure 1.5.
- c) Suggest possible reasons for the differences in unemployment rates in these countries. (After studying this unit, use the model of the labour market to expand on your answer to this question).

15. The table shows quarterly CPI data for Brazil in 2021–2022, taken from the World Bank's inflation database. Based on this data, read the following statements and choose the correct option(s).

Year and quarter	CPI value
2021 Q1	130.5

Year and quarter	CPI value
2021 Q2	133
2021 Q3	136.5
2021 Q4	140.9
2022 Q1	144.5
2022 Q2	148.9
2022 Q3	148.3
2022 Q4	149.4

- a) With 2021 Q1 as the base period, the 2022 Q1 CPI value would be 112.
- b) With 2021 Q1 as the base period, the 2022 Q4 CPI value would be 114.
- c) With 2022 Q1 as the base period, the 2021 Q1 CPI value would be 90.
- d) The purchasing power of people in Brazil has generally decreased over the period shown.
- 16.** The graph shows the economy at point A on the wage-setting curve, with 64 people employed (out of a population of 100). Based on the information in the graph, read the following statements and choose the correct option(s). All numbers given are rounded to the nearest whole number.



- a) The participation rate is 89%.
- b) The unemployment rate is 29%.
- c) The employment rate is 80%.
- d) If some inactive workers decided to enter the labour force, then *ceteris paribus*, the unemployment rate and participation rate would both increase.

4. CONCEPTUAL QUESTIONS

- 17.** We saw that if an economy has low aggregate demand with high cyclical unemployment, then automatic adjustment back to equilibrium could occur through a process of wage and price

cuts. Imagine you are a worker, and you see that many workers have lost their jobs while other workers are having their wages cut. How might this affect your spending and saving decisions?

18. Referring to Unit 6, provide a brief explanation of the shift in the wage-setting curve if a monitoring device to detect shirking is invented. Give an example from a real-world workplace of such a technology.